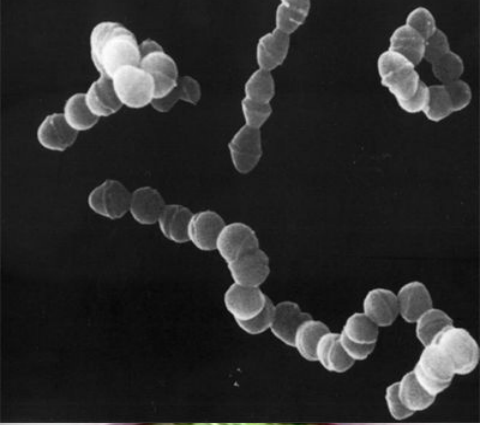


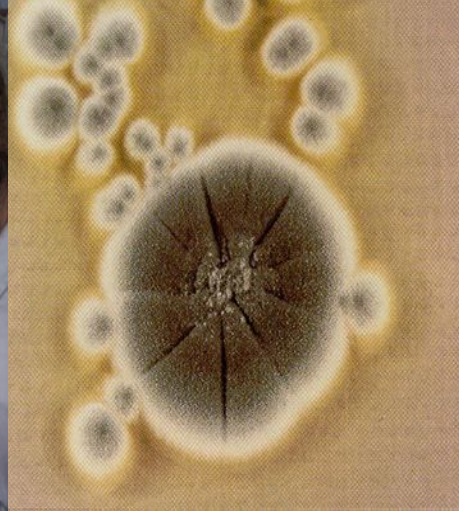
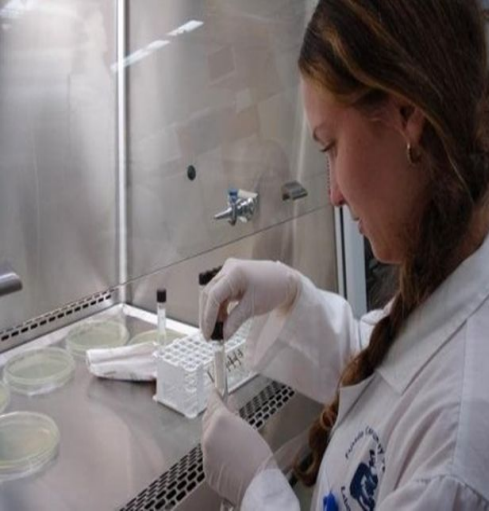


GENERAL MICROBIOLOGY

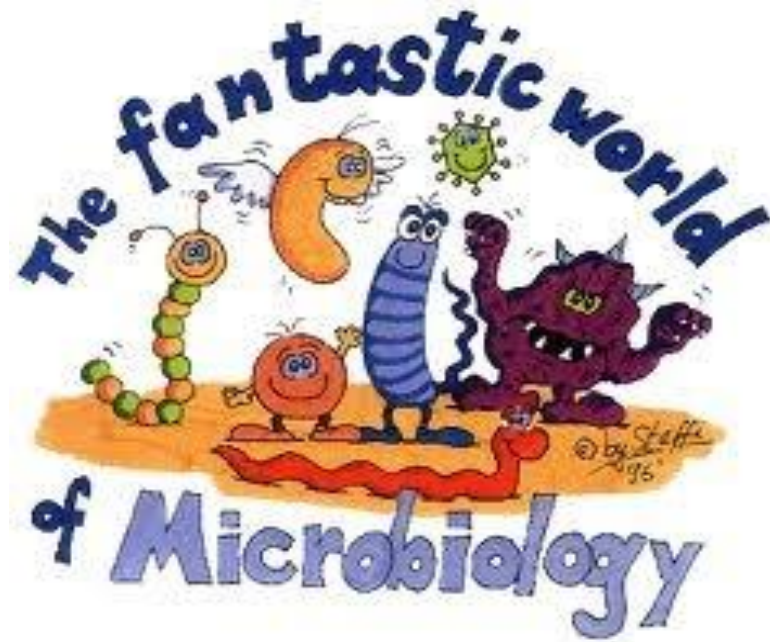
BIO 1203

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2021



Unifying Themes



The invisible world of microbes underlies and shapes what we call the “visible world”

Microbes have extraordinary genetic and metabolic diversity

Different microbes are adapted to survive and exploit an enormous range of environments, both live and dead.

Among all life forms on earth, microbes have the widest range of genetic and evolutionary diversity.

In addition to their ecological roles microbes are used industrially.

Microbiology is the study of microorganisms.

The overall theme of the **Microbiology** course is to study the relationship between microbes and our lives.

Microorganisms (microbes) are organisms that are too small to be seen with the unaided eye, and usually require a microscope to be seen.

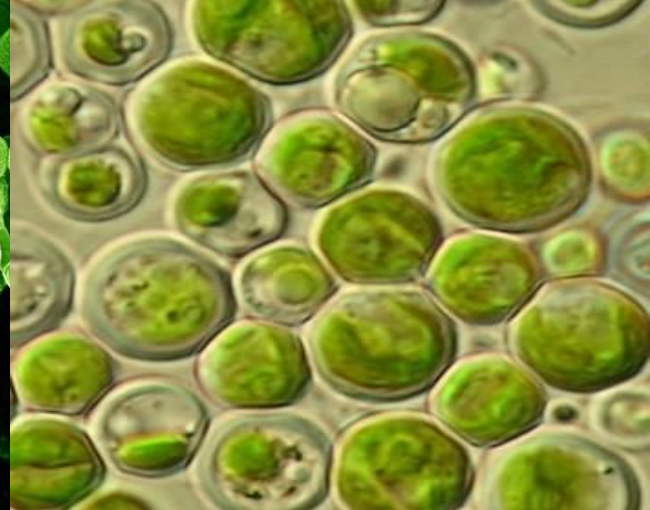
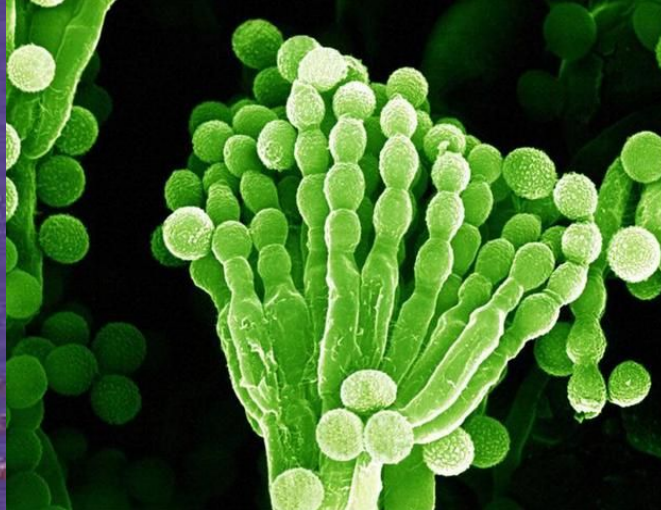
Micro - too small to be seen with the naked eye

Bio - life

ology - study of

This relationship involves harmful effects such as diseases and food spoilage as well as many beneficial effects.

“**Germ**” refers to a rapidly growing cell.



Microorganisms include:

Bacteria

Fungi (yeasts and molds)

Microscopic Algae

Protozoa

Viruses, Viroids, Prions

(Non-living infectious agents)

Microorganisms in our lives

These small organisms are usually associated with major diseases such as AIDS, uncomfortable infections, or food spoilage.

However, the majority of microorganisms make crucial contributions to the welfare of the world's inhabitants by maintaining balance of living organisms and chemicals in our environment.

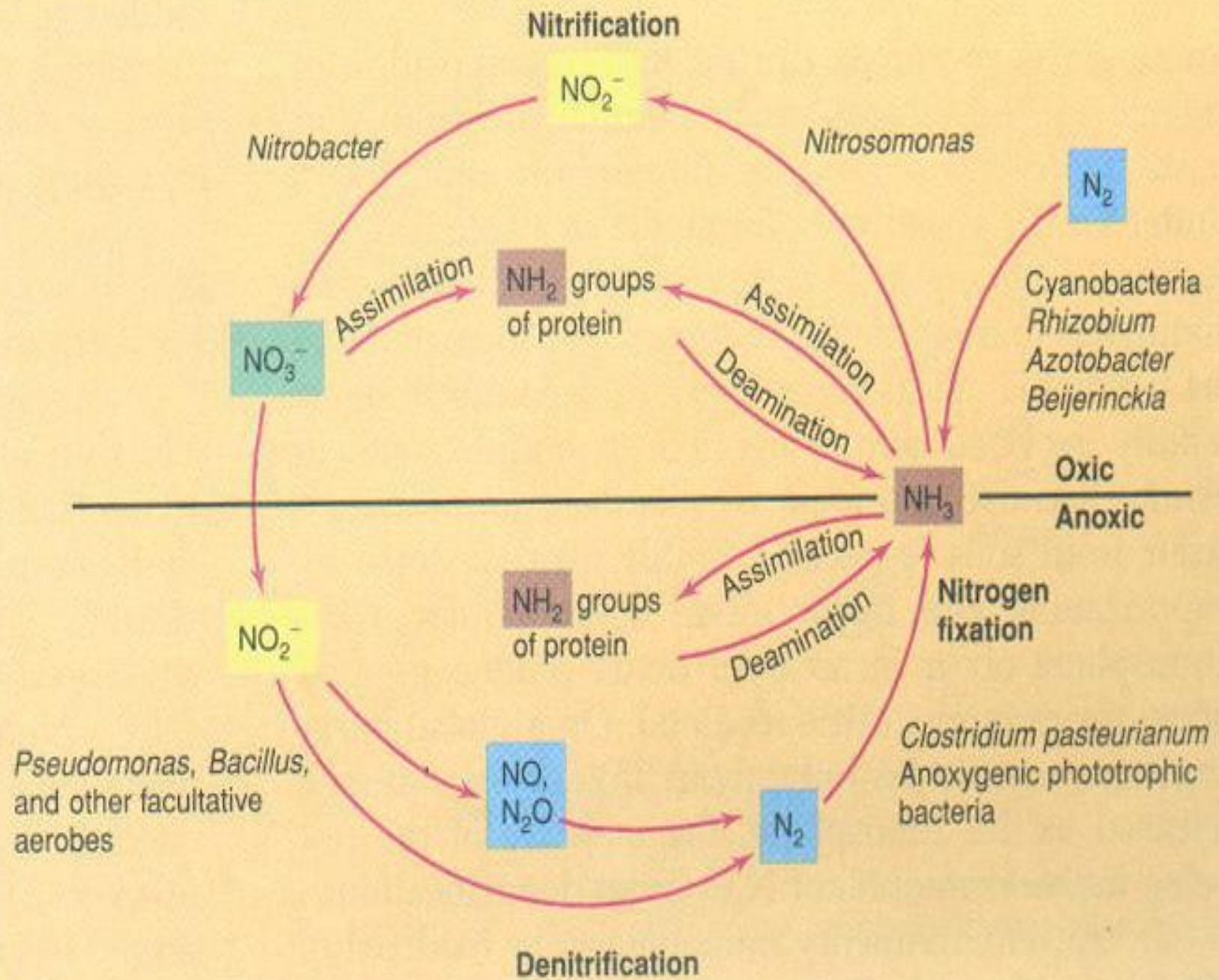
Therefore, Microorganisms are essential for life on earth.

They have important beneficial biological functions:

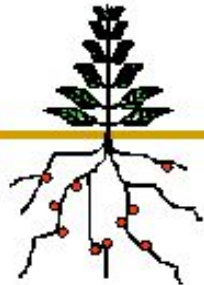
Photosynthesis: Marine and freshwater MO (Algae and some bacteria) capture energy from sunlight and convert it to food, forming the basis of the **food chain** in oceans, lakes, and rivers and **generates oxygen** which is critical for life on Earth.

Decomposers: Soil microbes break down dead and decaying matter and **recycle chemical elements** that can be used by other organisms.

Nitrogen Fixation: Some bacteria can take nitrogen from air and incorporate it into organic compounds in soil, water, and air.



Ecology of nitrogen-fixing bacteria

System of N ₂ fixation (and microbes involved) (N ₂ → NH ₃)	SYMBIOSIS (e.g. <i>Rhizobium</i>)	ASSOCIATION (e.g. <i>Azospirillum</i>)	FREE-LIVING (e.g. <i>Rhodospirillum</i>)
			
Energy source (Organic C)	Sucrose from the host plant	Root exudates from the host plant	Heterotroph (plant residues) Autotroph (photo-synthesis)
Estimates of fixation rate (kg N/ha/y)	50-400	10-200	1-2 10-80

Only minority of all MOs are pathogenic.

Microbes that cause food spoilage are also a minority.

The vast majority of microbes benefit humans, other animals, and plants in many ways.

RECYCLING VITAL ELEMENTS

In 1880s, **Beijerinck** and **Winogradsky** showed how bacteria help recycle vital elements between the soil and the atmosphere.

Microbial ecology: the study of the relationship between microorganisms and their environment.

Microorganisms recycle carbon, nitrogen, sulfur, oxygen, and phosphorus into forms that can be used by plants and animals.

Bacteria and fungi, return CO₂ to the atmosphere when decomposing organic wastes and dead plants and animals.

Algae, cyanobacteria, and plants use CO₂ to produce carbohydrates.

Digestion: Human and many other animals have microorganisms in their digestive tract, that are essential for digestion and vitamin synthesis.

Cellulose digestion by ruminants (cows, rabbits, etc.)

Synthesis of Vitamin K (for blood clotting) and Vitamin B (for metabolism) in humans.

Synthesis of chemical products: MOs have many commercial applications, such as the synthesis of acetone, organic acids, enzymes, alcohols.

What can we do with knowing microbiology

There are large amount of different applications in Microbiology

Some of those are,

- Biomedical science

- Pharmaceutical microbiology

- Industrial microbiology

- Microbial biotechnology

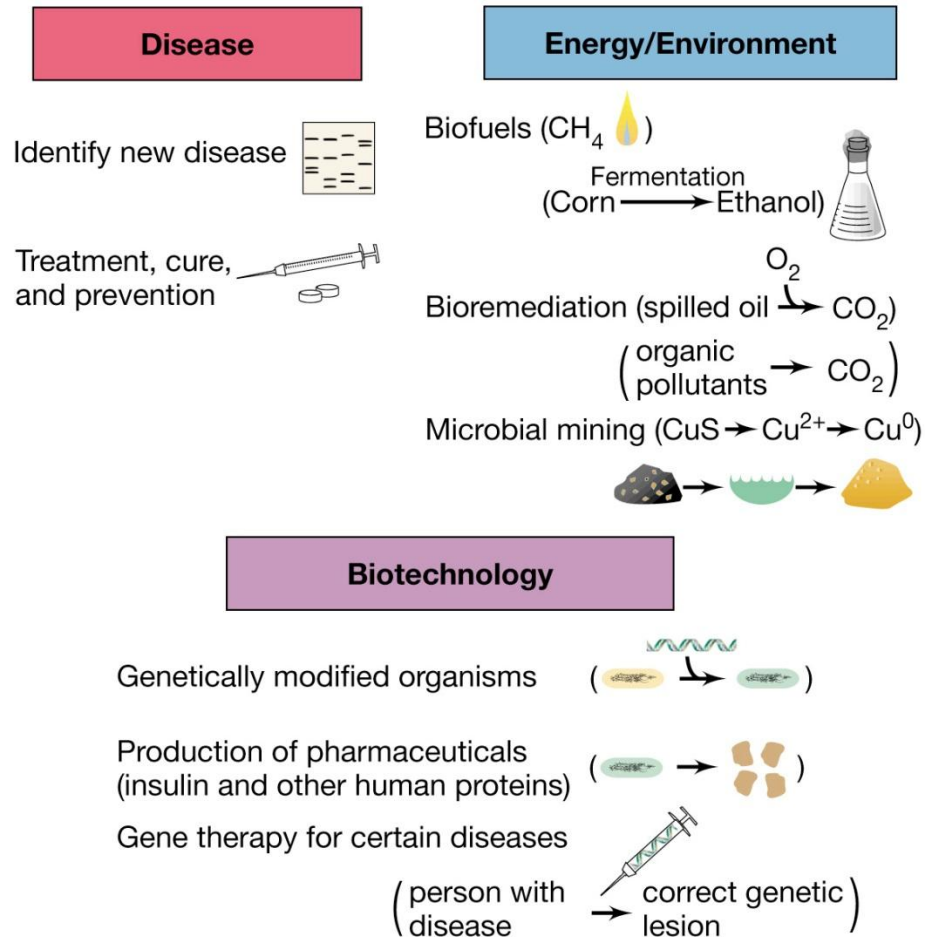
- Food microbiology

- Agricultural microbiology

- Environmental microbiology

- Molecular Biology

Use of Microorganisms





Food Microbiology



The Roles of Microorganisms in Food Production

Use starter cultures in commercial food and beverage production

- Composed of known microorganisms

- Consistently perform specific fermentations

- Many common products result from fermentation of vegetables, meats, and dairy products

Products of alcoholic fermentation

Microorganisms convert simple sugars into alcohol and carbon dioxide

Specific starter cultures used in commercial applications of alcohol fermentation

Various alcoholic products made through fermentation



Food industry: many important foods and beverages are made with microbes: vinegar, pickles, alcoholic beverages, green olives, soy sauce, buttermilk, cheese, yogurt, and bread.

Curd: solid casein from lactic acid bacteria and rennin

Hard cheeses produced by lactic acid bacteria

Semisoft cheeses ripened by *Penicillium* on surface

Lactobacillus acidophilus The bacterium is a member of the normal flora of humans, found in the oral cavity, the small intestine

Where it is thought to play a beneficial role. The organism is generally the first bacterium listed as present in probiotic food



Elmer L. Gaden, Jr.

(a)



Elmer L. Gaden, Jr.

(b)

Agricultural microbiology

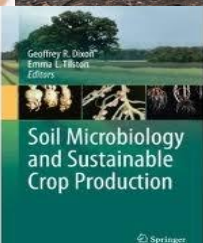
Organic farming

Microbial inoculums

Compost and
biofertilizers

Crop plant pathology

Microbial insecticides
and pesticides





If

People follow conventional agricultural methods

Heavy usage of chemical fertilizers

Produce high yield

Leads to environmental pollution & destruction of useful soil microbes

As an alternative

Can use organic & bio fertilizers

No pollution

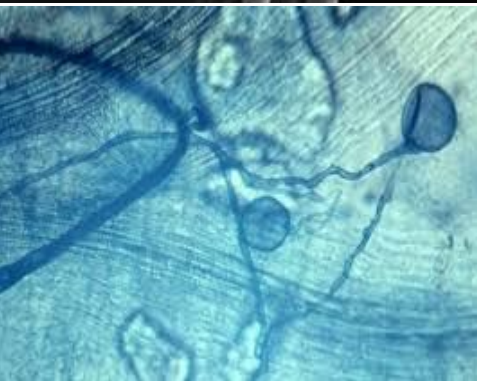
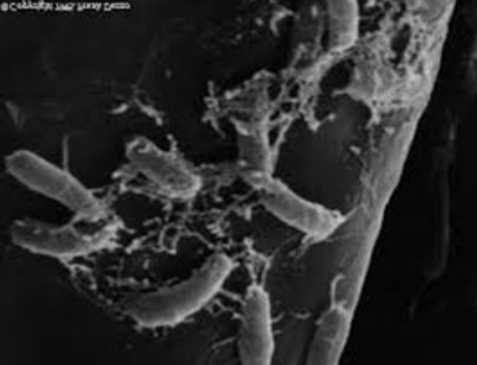
Environment friendly & sustainable

Why do we need biofertilizers?

Plant growth promoting bacteria and fungi

Supply nutrients to crops, stimulate plant growth, e.g., through the production of plant hormones, control or inhibit the activity of plant pathogens and improve soil structure

Therefore, the effective use of organic materials and beneficial microbial inoculants in farming is likely to be promoted



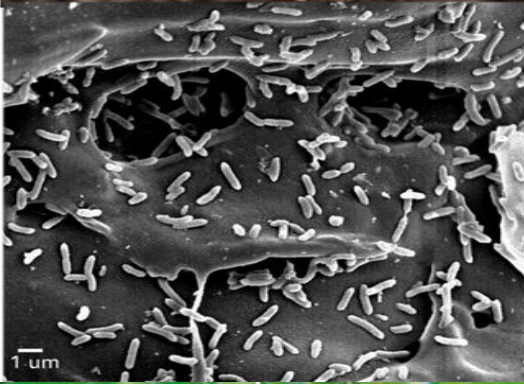


Pesticides and agricultural products



Microbes used to help crop management

Biosensors and bioreporters



Use of microorganisms to solve environmental problems

Biosensors

Bacteria or microbial products combined with electronic measuring devices

Bioreporters

Composed of microbes with innate signaling capabilities





What are we going to do with waste?

Dumping

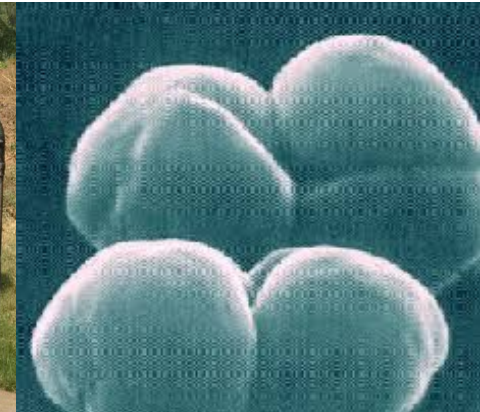
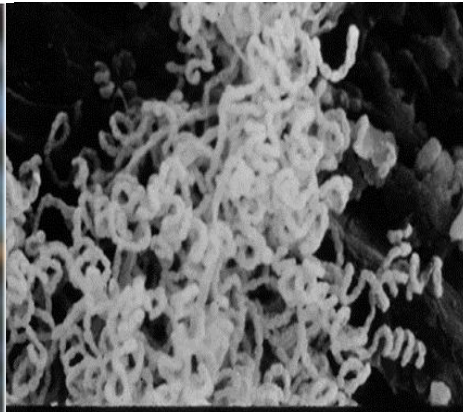
Recycling

Producing something useful

e.g.

Compost

Biogas (CH_4)





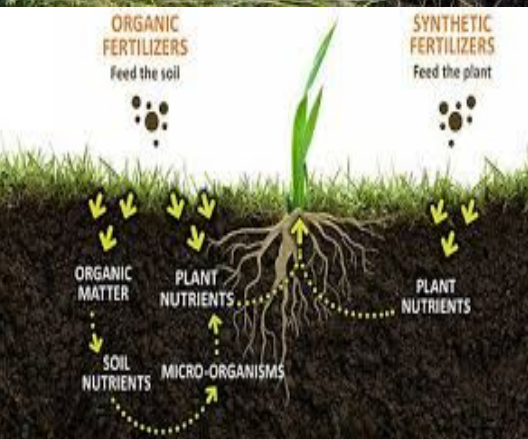
Principles of Composting

What Is Compost?

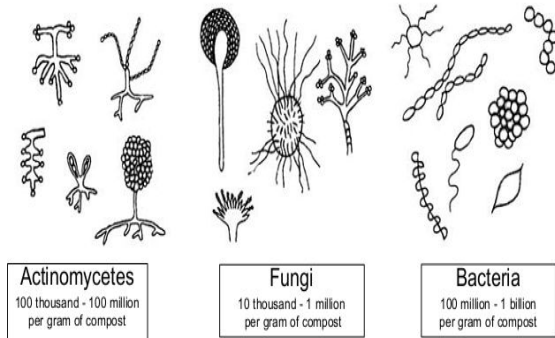
The product resulting from the controlled biological decomposition of organic materials

Sanitized through the generation of heat
Stabilized to the point where it is beneficial to plant growth

Provides humus, nutrients, and trace elements to soils



COMPOST MICROORGANISMS MAGNIFIED 1,000 TIMES



Microbes break down organic compounds to:

- Obtain energy to carry on life processes
- Acquire nutrients (N, P, K) to sustain populations

Aerobic respiration most efficient process in terms of energy yield.

Aerobic breakdown of organic materials is quickest way to achieve biological stability.

“Heat” generated in composting is byproduct of microbial metabolism.

Water Treatment

Water pollution

Water pollution can occur three ways

Physically, Chemically, Biologically

Polluted waters support a greater than normal microbial load

SEWAGE TREATMENT: Using microbes to recycle water.

Recycling water and prevent the pollution of rivers and oceans

Bacteria degrade organic matter in sewage (99% water), producing such by-products as carbon dioxide, nitrates, phosphates, sulfates, ammonia, hydrogen sulfide, and methane

Waterborne illnesses

Consuming contaminated water can cause various diseases

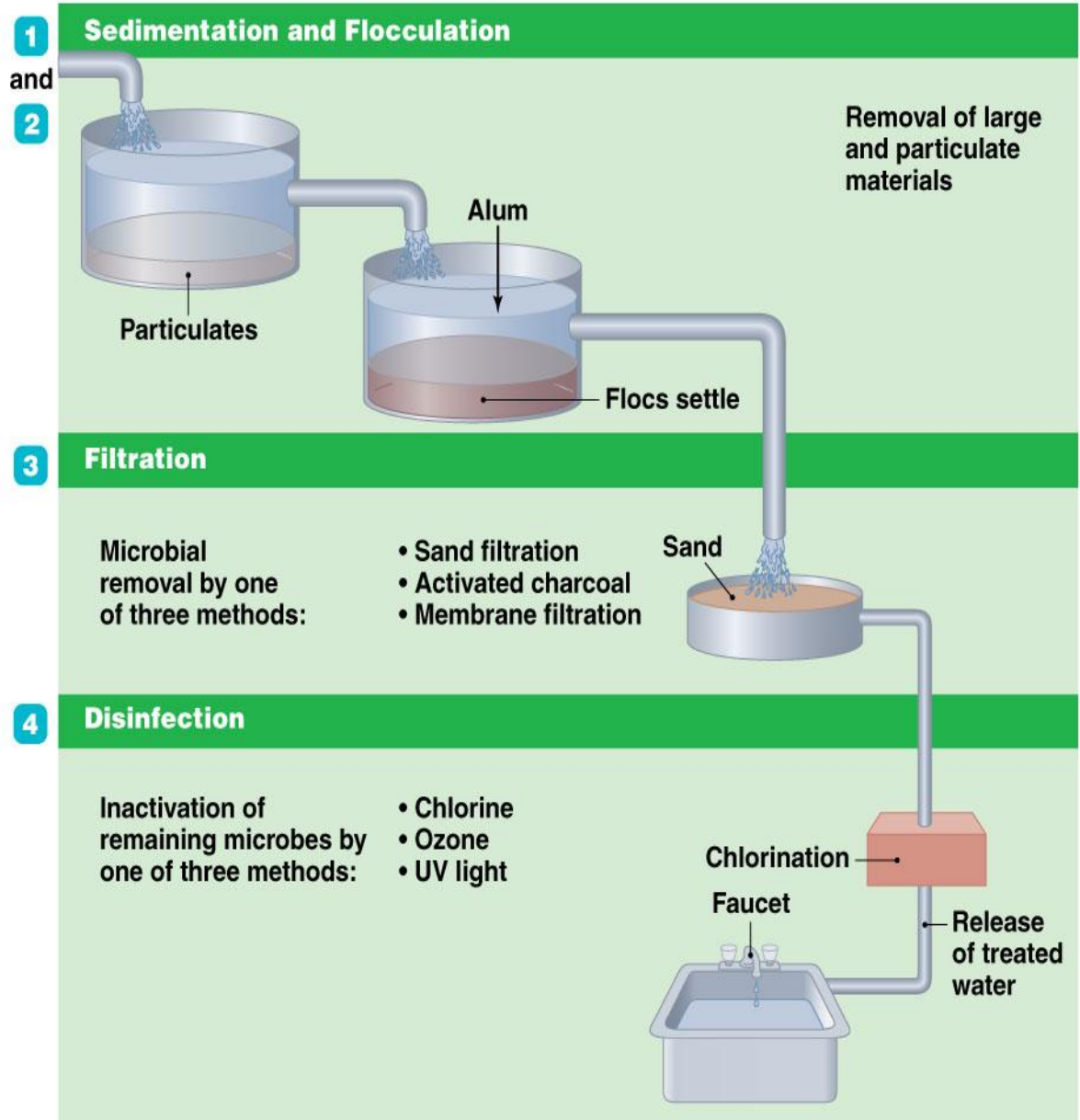
Diarrheal diseases occur worldwide

Outbreaks are point-source infections

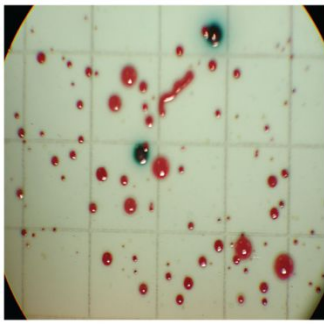
Water treatment removes most waterborne pathogens



(a)



(b)



(a)
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(b)

Water Treatment

Water quality testing

Majority of waterborne illnesses caused by fecal contaminated water

Indicator organisms signal possible presence of pathogens

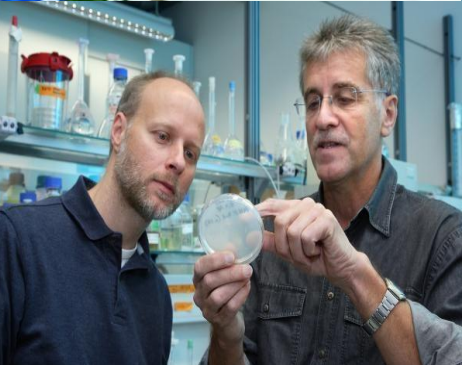
E. coli or other coliforms used as indicator organisms

E. coli is a good indicator organism

Consistently found in human waste

Survives in water as long as most pathogens

Easily detected by simple tests



Microbial Pharmaceuticals

Medicine: Many **antibiotics** and other drugs are naturally synthesized by microbes. Penicillin is made by a mold.

Genetic engineering: recombinant microbes produce important Medical and therapeutic products: human growth hormone, insulin, blood clotting factor, recombinant vaccines, monoclonal antibodies,...etc.

Medical Research: Microbes are well suited for biological and medical research for several reasons:

Relatively simple and small structures, easy to study

Genetic material is easily manipulated.

Can grow a large number of cells very quickly and at low cost. Short generation times make them very useful to study genetic changes. **Though only a minority of MOs are pathogenic (disease-producing), practical knowledge of microbes is necessary for medicine and related health sciences.**

Bioremediation

Uses organisms to clean up toxic, hazardous, or recalcitrant compounds by degrading them to harmless compounds

Most known application is use of bacteria to clean oil spills

Such bioremedial microbes are *Pseudomonas* and *Bacillus*,

Two Types of Bioremediation

Natural bioremediation

Microbes “encouraged” to degrade toxic substances in soil or water

Addition of nutrients stimulate microbe growth

Artificial bioremediation

Genetically modified microbes degrade specific pollutants



Industrial Products of Microorganisms

Enzymes and other industrial products

Microbial products used as food additives and supplements

Include vitamins, amino acids, organic acids, dyes

Alternative fuels

Some microbes produce carbohydrates used as fuels

Other microbes convert biomass into renewable fuels

Pharmaceuticals

Includes antimicrobials, recombinant hormones, and other cell regulators

Biological Warfare and Bioterrorism

Microbes can be fashioned into biological weapons

Bioterrorism- Uses microbes or their toxins to terrorize human populations

Agroterrorism -Uses microbes to terrorize human populations by destroying the food supply

Animal pathogens

Divided into categories based on level of danger

Some agents could potentially amplify an outbreak

Infect wild animal populations in addition to livestock

Foot-and-mouth disease is most dangerous of the agents

It affects all wild and domestic animals

Plant pathogens

Most potential agents are fungi

Dissemination could easily result in contamination of soil

All agents are naturally present

Detecting difference between a natural outbreak and and intentional attack would be difficult

INSECT PEST CONTROL BY MOs

Insect pest control is important for both agriculture and the prevention of human diseases.

Bacillus thuringiensis infections are fatal for many insects but harmless to other animals, including humans, and to plants.

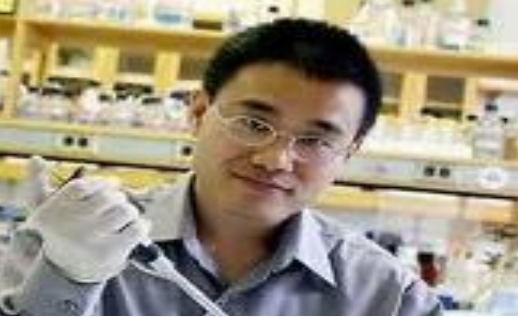
The bacteria produce protein crystals that are toxic to the digestive systems of the insects.

The toxin gene has been inserted into some plants to make them insect resistant.

Microbes that are pathogenic to insects are alternatives to chemical pesticides in preventing insect damage to agricultural crops, disease transmission, and avoid harming the environment.



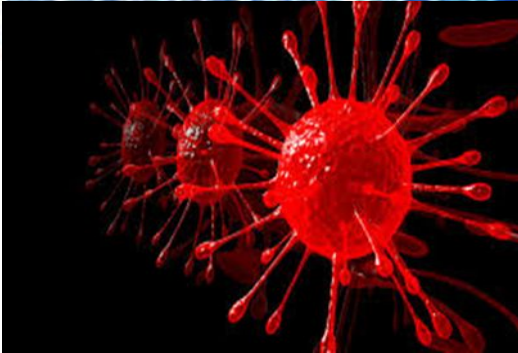
Microbial biotechnology



Manipulation of microbes at the genetic and molecular level to generate useful products



Bioengineered microbes for different industrial processes



MODERN BIOTECHNOLOGY AND RECOMBINANT DNA TECHNOLOGY

Biotechnology, the use of microbes to produce foods and chemicals, is centuries old.

Genetic engineering is a new technique for biotechnology. Through genetic engineering, bacteria and fungi can produce a variety of proteins including vaccines and enzymes.

Recombinant DNA techniques have been used to produce a number of natural proteins, vaccines, and enzymes.

Genetically modified bacteria are used to protect crops from insects, from freezing, and to improve the appearance, flavor, and shelf life of fruits and vegetables. (more: Drought resistance and temperature tolerance)

Microbes and human diseases

We all live in a world filled with microbes, and we all have a variety of microorganisms on and in our bodies.

Microbes normally present in and on the human body are called normal microbiota, or microflora.

Bacteria were once classified as plants giving rise to use of the term flora for microbes.

This term has been replaced by microbiota.

The normal microbiota not only harmless, but also benefit us.

Some protect us against disease by preventing the over-growth of harmful microbes.

Others produce useful substances such as vitamin K and B.

Unfortunately, under some circumstances normal microbiota can make us sick or infect people we contact.

Microbes and human diseases

An infectious disease is one in which pathogens invade a susceptible host, such as a human or animal.

The pathogen carries out at least part of its life cycle inside the host, and disease frequently results.

When a pathogen overcomes the host's resistance, disease results.

Many mistakenly believed that infectious diseases were under control

Malaria would be eradicated by killing mosquitoes by DDT.

A vaccine would prevent diphtheria.

Improved sanitation measures would help prevent cholera transmission.

Recent outbreaks of such infectious diseases indicates that not only they are not disappearing, but seem to be reemerging and increasing. In addition, a number of new diseases -Emerging infectious diseases (EID)-have cropped up in recent years

Microbes and human diseases

- Emerging infectious diseases (EID): are diseases that are new or changing and are increasing or have the potential to increase in incidence in the near future.
- Some factors that have contributed to the emergence of EIDs:
 - a. Evolutionary changes in existing organisms.
 - b. The spread of known diseases to new geographic regions or populations by modern transportation.
 - c. Increased human exposure to new, unusual infectious agents.

1. West Nile encephalitis

Caused by West Nile virus

First diagnosed in the West Nile region of Uganda in
1937

Appeared in New York City in 1999

2. Bovine spongiform encephalopathy

- a. Caused by prion

- b. Also causes Creutzfeldt-Jakob disease (CJD)

- c. New variant CJD in humans is related to cattle feed
from infected sheep.

Emerging Infectious Diseases

3. *Escherichia coli* O157:H7

- a. Toxin-producing strain of *E. coli*
- b. First seen in 1982
- c. Leading cause of diarrhea worldwide

4. *Ebola* hemorrhagic fever

- a. Caused by Ebola virus
- b. Causes fever, hemorrhaging, and blood clotting
- c. First identified near Ebola River, Congo
- d. Outbreaks every few years.

5. Invasive group A *Streptococcus*

- a. Rapidly growing bacteria that cause extensive tissue damage

6. Avian influenza A (H5N1) Caused by Influenza A virus (H5N1)
- a. Primarily in waterfowl and poultry
 - b. Sustained human-to-human transmission has not occurred yet

7. Severe acute respiratory syndrome (SARS)

- a. SARS-associated Coronavirus
- b. Occurred in 2002-2003
- c. Person-to-person transmission

8. Cryptosporidiosis

- a. Caused by *Cryptosporidium* protozoa
- b. First reported in 1976
- c. Causes 30% of diarrheal illness in developing countries

9. Acquired immunodeficiency syndrome (AIDS)

- a. Caused by Human immunodeficiency virus (HIV)
- b. First identified in 1981
- c. Worldwide epidemic infecting 44 million people; 14,000 new infections daily
- d. Sexually transmitted disease affecting males and females